

Python Conversational Loop – 30-Minute Task

Assignment: Build Your Own Rule-Based Python Chatbot

1. Aim

To develop a simple rule-based conversational chatbot in Python that continuously interacts with the user, recognizes different messages, provides suitable responses, and allows the user to exit.

2. Objectives

- Use `input()` to receive user messages.
- Use `while True` for continuous conversation.
- Clean input using `.lower().strip()`.
- Recognize greetings such as hello and hi.
- Answer at least three different questions or topics.
- Use `if / elif / else` for decision making.
- Use `break` to exit when the user enters bye or exit.
- Use the user's name in at least one response.
- Provide a useful response for unknown messages.
- Use the `in` operator so variations such as hello bot can trigger a greeting.

3. Algorithm

Step 1: Start the program.

Step 2: Ask the user to enter their name.

Step 3: Display a welcome message using the user's name.

Step 4: Start an infinite loop using `while True`.

Step 5: Ask the user to enter a message.

Step 6: Clean the message using `lower()` and `strip()`.

Step 7: Check for bye or exit and terminate using `break`.

Step 8: Check greetings and different topics using `if/elif` conditions.

Step 9: Use `else` for unknown messages.

Step 10: Repeat the conversation until the user exits.

Step 11: End the program.

4. Python Program

```
print("Welcome to Python Chatbot!")

name = input("What is your name? ").strip()

print("Hello", name, "! Let's start our conversation.")

while True:
    message = input("You: ").lower().strip()

    if "bye" in message or "exit" in message:
        print("Bot: Goodbye", name, "! Have a nice day!")
        break

    elif "hello" in message or "hi" in message:
        print("Bot: Hello", name, "! How can I help you?")

    elif "python" in message:
        print("Bot: Python is a popular programming language.")

    elif "study" in message:
        print("Bot: Regular practice and revision can improve your studies.")

    elif "your name" in message:
        print("Bot: I am a simple Python rule-based chatbot.")

    else:
        print("Bot: Sorry, I don't understand that. "
              "Try asking about Python, study, or my name.")
```

5. Code Explanation

input(): Receives input from the user.

.lower().strip(): Converts the message to lowercase and removes unnecessary spaces.

while True: Keeps the conversation running continuously.

if / elif / else: Makes decisions based on the user's message.

in: Allows variations such as hello bot or hey hello to be recognized.

break: Stops the loop when the user wants to exit.

6. Sample Output

```
Welcome to Python Chatbot!
What is your name? Rajitha
Hello Rajitha! Let's start our conversation.

You: hello
Bot: Hello Rajitha! How can I help you?

You: what is python
Bot: Python is a popular programming language.

You: I want to study
Bot: Regular practice and revision can improve your studies.

You: what is your name
Bot: I am a simple Python rule-based chatbot.

You: random question
Bot: Sorry, I don't understand that. Try asking about Python, study, or my name.
```

You: bye
Bot: Goodbye Rajitha! Have a nice day!

7. Test Cases

Test Input	Expected Result
hello	Greeting response
HELLO	Greeting response
hi	Greeting response
hello bot	Greeting response
hey hello	Greeting response
What is Python?	Python information
I want to study	Study advice
What is your name?	Chatbot introduction
random question	Unknown-message response
bye / exit	Goodbye and program exits

8. Concepts Used

Python Concept	Purpose
input()	Gets input from the user
print()	Displays responses
while True	Keeps the chatbot running
.lower()	Converts input to lowercase
.strip()	Removes extra spaces
if / elif / else	Decision making
in	Checks for words inside a message
break	Exits the loop

9. Conclusion

The rule-based Python chatbot was successfully developed using a conversational loop. It accepts user input, processes messages, recognizes greetings and different topics, provides appropriate responses, handles unknown messages, and terminates when the user enters bye or exit. The overall flow is INPUT → PROCESS → RESPOND → REPEAT → EXIT.